

Comprehensive Analysis of Auxiliary Electric Motors in European and Asian Battery Electric Vehicles (2024-2026 Models)

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1. Executive Summary

This report presents a comprehensive and in-depth analysis of the non-drivetrain auxiliary electric motors utilized in Battery Electric Vehicles (BEVs) from European and Asian manufacturers, specifically focusing on models sold within the European market for the 2024 to 2026 model years. The research delineates these critical components into three distinct categories based on power and design: small stepper motors with a power rating under 50 watts, medium DC motors rated between 50 and 500 watts, and medium stepper motors with a power rating of 50 to 200 watts. The investigation spans the full spectrum of passenger vehicle segments, from A-segment mini cars to F-segment luxury flagships, to provide a holistic view of motor distribution and complexity across the market.

The key findings of this report underscore the dramatic proliferation and increasing sophistication of auxiliary motors as a direct consequence of vehicle electrification and the rising consumer demand for advanced comfort, convenience, and safety features. The absence of a traditional internal combustion engine necessitates the complete electrification of all ancillary systems, making the efficiency and reliability of these motors paramount to overall vehicle performance and, most critically, driving range. Small stepper motors are identified as the cornerstone of precision control systems, finding ubiquitous application in Heating, Ventilation, and Air Conditioning (HVAC) blend door actuators and adaptive headlamp leveling systems, where their ability for precise, open-loop positioning is invaluable [1, 3, 6, 7]. Medium DC motors, particularly of the brushless (BLDC) variety, have emerged as the workhorses for continuous-duty applications, most significantly driving the electric water pumps essential for the complex thermal management of the high-voltage battery and powertrain components [21, 22, 88, 93].

A central finding of this analysis is the direct and exponential correlation between a vehicle's market segment and its total count of auxiliary motors. Economical A-segment vehicles may contain as few as nine motors, whereas high-end F-segment luxury vehicles can easily exceed eighty-five individual motors, driven by features such as multi-zone climate control, executive-class power seats with massage functions, and active aerodynamic systems. The category of medium stepper motors (50-200W) was found to be largely absent from common auxiliary applications, suggesting its use is confined to highly specialized or niche systems not widely documented.

Furthermore, the report delves into the material composition and supply chain dynamics that underpin the production of these motors. The analysis reveals a critical dependence on specific raw materials, namely high-conductivity copper for windings and high-grade electrical steel for stator cores [21, 22, 48]. For high-performance motors, the use of Neodymium Iron Boron (NdFeB) rare earth permanent magnets is prevalent, which introduces significant cost volatility and profound geopolitical supply chain risk [74, 81]. The global supply chain for these magnets is overwhelmingly dominated by China, creating a systemic vulnerability for the global automotive industry [81]. In response, automakers and governments are actively pursuing mitigation strategies, including supply chain diversification, vertical integration through upstream investments in mining and processing, and research into alternative mo-

tor technologies [81, 82, 84]. These material and supply chain challenges are identified as defining factors that will shape the cost, availability, and technological trajectory of auxiliary motors and the broader BEV market in the coming years.

2. Introduction

The global automotive industry is undergoing a paradigm shift, with the transition to battery electric vehicles (BEVs) representing the most significant technological evolution in over a century. This transformation extends far beyond the primary drivetrain, fundamentally re-engineering the entire vehicle architecture. A crucial and often underappreciated aspect of this evolution is the comprehensive electrification of auxiliary systems. In a conventional vehicle powered by an internal combustion engine (ICE), numerous functions such as power steering, climate control, and vacuum-assisted braking derive their power mechanically, hydraulically, or pneumatically from the running engine. In a BEV, every one of these functions must be independently powered by its own electric motor. This has resulted in the integration of a vast and diverse array of small and medium-sized electric motors, each performing a specialized task critical to vehicle safety, operational efficiency, and passenger comfort. The cumulative energy consumption of these auxiliary systems has a direct and measurable impact on the vehicle's total energy budget, making the efficiency, weight, and control of their motors a primary concern for engineers striving to maximize driving range.

The scope of this report is to provide a detailed and focused investigation into these non-drivetrain auxiliary electric motors. The analysis is specifically tailored to European and Asian manufactured BEV models projected for sale in the European market during the 2024-2026 model years. The research methodology involves the synthesis of data from a wide range of sources, including automotive industry technical documentation, component manufacturer specifications, academic papers, and specialized industry publications. This approach allows for a multi-faceted view of the subject, combining theoretical principles with real-world application data. The report is structured to provide a clear and logical progression of information, beginning with the fundamental types of motors and their applications, moving to their distribution across vehicle classes, and culminating in a deep-dive analysis of their material composition and the complex global supply chains that support their production.

For the purpose of this analysis, auxiliary motors have been classified into three distinct categories based on their power rating and fundamental operating principle. The first category is **small stepper motors**, defined as having a power output of less than 50 watts. These motors are characterized by their ability to rotate in precise, discrete angular increments, making them ideal for positioning tasks. The second category is **medium DC motors**, defined by a power rating between 50 and 500 watts. This category, which includes both brushed and brushless (BLDC) designs, is primarily used for applications requiring continuous rotational power, such as pumps and fans. The third category is **medium stepper motors**, with a power range of 50 to 200 watts, representing a higher-torque variant of the stepper motor for more demanding positioning applications.

To understand how the application and quantity of these motors vary with vehicle price and complexity, the report examines their distribution across the standard European vehicle segments. These segments provide a framework for comparing vehicles of similar size and market position. The segments covered in this report are:

- * **A-segment:** Mini cars (e.g., Fiat 500e)
- * **B-segment:** Small cars (e.g., Peugeot e-208, VW ID.3)
- * **C-segment:** Compact cars (e.g., Tesla Model 3, Nissan Ariya)
- * **D-segment:** Mid-size cars (e.g., BMW i4, Polestar 2)
- * **E-segment:** Executive cars (e.g., Mercedes EQS, Audi e-tron GT)
- * **F-segment:** Luxury cars (e.g., Porsche Taycan, Mercedes EQS SUV)

By examining the interplay between motor type, application, vehicle segment, and material science, this report aims to provide a comprehensive and authoritative overview for industry stakeholders, engineers, and analysts seeking to understand the component-level architecture and strategic dependencies of modern battery electric vehicles.

3. Auxiliary Motor Types, Locations, and Applications

The functionality of a modern battery electric vehicle is dependent on a distributed network of auxiliary electric motors, each selected and engineered for a specific task. The choice between motor types is a critical engineering decision, balancing requirements for precision, power, efficiency, weight, and cost. This section provides a detailed description of the three primary motor categories investigated in this report, their typical in-vehicle locations, and their principal applications.

Small Stepper Motors (<50 Watts)

Small stepper motors, with power ratings consistently below 50 watts, are foundational components for a multitude of precision control systems within BEVs. Their defining operational characteristic is the ability to rotate in discrete, fixed angular increments, known as “steps.” This allows for exceptionally accurate positioning of mechanical components without the need for external feedback sensors like encoders or resolvers. This capability, referred to as **open-loop control**, is a significant advantage, as it reduces system complexity, cost, and potential points of failure [3, 7]. In the context of BEVs, where every watt of energy consumed can affect the vehicle’s driving range, the inherent efficiency and low standby power consumption of these motors are highly valued. They are predominantly found in systems that require the exact and repeatable positioning of components such as flaps, valves, mirrors, and lenses. Control is typically managed by dedicated microcontrollers or driver ICs that receive commands over vehicle data networks, such as the Local Interconnect Network (LIN) bus, enabling sophisticated and coordinated operation of multiple actuators [6, 7].

A primary and ubiquitous application for small stepper motors is within the vehicle’s Heating, Ventilation, and Air Conditioning (HVAC) system [1, 3, 4]. Located deep within the main HVAC module, which is typically situated behind the dashboard and firewall, these motors function as actuators that drive the various doors and flaps controlling the flow and temperature of air within the cabin. A critical function is the operation of the **blend door**, which precisely mixes hot air from the heater core (or heat pump) with chilled air from the evaporator [3, 4]. The stepper motor allows the climate control unit to position this door with extreme accuracy, enabling the system to achieve and maintain the exact temperature selected by the occupants. Research has shown that the use of stepper motors in this role can significantly reduce the energy consumption of the actuator system compared to older DC motor designs, a crucial benefit in an electric vehicle [3, 4]. Beyond temperature, stepper motors also control the **mode doors**, which direct airflow to the various cabin outlets—such as the windscreen defroster, dashboard face vents, and floor-level footwell vents—and the **recirculation flap**, which toggles the system’s air intake between fresh outside air and recirculated cabin air. The silent, precise, and reliable operation of these motors is a key contributor to the perceived quality and comfort of the vehicle’s interior environment.

Another critical safety-related application for small stepper motors is within advanced headlamp systems [6, 7]. Modern BEVs are frequently equipped with automatic headlamp leveling and Adaptive Front-lighting Systems (AFS), both of which depend on stepper motors for their rapid and precise adjustments. These motors are housed directly within the headlamp assemblies themselves [7]. European regulations mandate automatic headlamp leveling for high-intensity light sources like HID xenon and powerful LEDs to prevent dazzling oncoming traffic [6, 7]. This system uses vehicle level sensors, often mounted on the front and rear suspension, to detect changes in the vehicle’s pitch

caused by passenger load, cargo, or dynamic driving events like acceleration and braking. A control unit processes this sensor data and commands the stepper motors to make minute vertical adjustments to the headlamp angle, ensuring the light beam remains parallel to the road surface [6, 7]. In more sophisticated Adaptive Front-lighting Systems, stepper motors also provide horizontal (swivel) adjustments. AFS utilizes real-time data from the vehicle's CAN bus, including steering angle and speed, to pivot the headlamps into a turn, proactively illuminating the road ahead rather than the roadside [7]. This dynamic lighting significantly improves driver visibility and reaction time during nighttime cornering. Each headlamp in an AFS typically contains two separate stepper motors: one for vertical leveling and another for horizontal swiveling [7]. These motors are engineered for extremely quiet, low-resonance operation, often using micro-stepping control techniques to achieve smooth, imperceptible beam movement.

Beyond these major applications, small stepper motors and functionally similar small DC motors are used in a variety of other low-power actuator roles. One such example is the **charging port door actuator** [16, 18]. This motorized mechanism locks and unlocks the flap covering the vehicle's charging port, integrating with the central locking system to secure the port when the vehicle is parked and provide access for charging. Another increasingly common application is in **active grille shutters** [11]. These systems feature motorized louvers in the front grille that are controlled by small actuators. The louvers automatically close at higher speeds to reduce aerodynamic drag and improve efficiency, and open when low-speed cooling is required for the battery or power electronics. By precisely managing airflow, these small actuators make a direct contribution to optimizing the vehicle's energy consumption and extending its driving range.

Medium DC Motors (50 - 500 Watts)

Medium-power Direct Current (DC) motors, and particularly their more advanced brushless (BLDC) variants, serve as the workhorses in BEV auxiliary systems that demand continuous operation and significantly higher power output than small actuators can provide. This category, defined for this report as motors with a power rating between 50 and 500 watts, is indispensable for fluid-pumping and air-moving applications. Unlike stepper motors, which are optimized for precise positioning, DC motors are designed for sustained rotational speed and torque, making them the ideal choice for driving pumps and fans. In a BEV, the absence of a belt-driven accessory drive from a combustion engine means that all thermal management functions must be fully electrified. Efficient and reliable thermal management is not merely a matter of passenger comfort; it is fundamental to the health, performance, safety, and longevity of the high-voltage battery, the power electronics, and the electric drive motor itself.

The most significant and critical application for medium DC motors in BEVs is driving the **electric water pumps** that circulate liquid coolant throughout the vehicle's various thermal management loops. A modern BEV often features multiple, independent cooling circuits that must be managed with precision. A primary loop is dedicated to maintaining the high-voltage battery pack within its optimal operating temperature window, typically between 20°C and 40°C. Overheating can lead to accelerated degradation and safety risks, while operating at too low a temperature can severely limit power output and charging speed. Another separate loop may be responsible for managing the heat generated by the power inverter and the main electric drive motor. A third circuit, often integrated with a sophisticated heat pump system, manages cabin heating and cooling, scavenging waste heat from the powertrain when possible to improve overall efficiency. Each of these circuits requires at least one, and sometimes more, dedicated electric water pump to ensure the continuous and controlled flow of coolant. These pumps are typically located in the front of the vehicle, near the components they service, or are integrated into larger, modular thermal management assemblies. The power consumption of these pumps is a critical design parameter, with typical ratings falling squarely within the 50-500

watt range, depending on the required flow rate and pressure of the specific circuit [21, 22]. The use of efficient BLDC motors is standard in this application due to their long lifespan, high reliability, and precise speed control, which allows the vehicle's thermal management system to dynamically adjust coolant flow based on real-time thermal loads, thereby optimizing energy consumption.

In addition to pumps, medium DC motors power a host of other familiar and essential vehicle functions. **Power window lift** mechanisms in each door use a geared DC motor to raise and lower the glass. The front **windshield wiper system** relies on a powerful DC motor, often with a complex linkage, to provide the force needed to clear rain and snow. In higher-trim vehicles, **power seat adjustment** is accomplished by a series of medium DC motors; a typical 8-way adjustable seat will have separate motors for fore/aft movement, front height, rear height, and backrest recline. More advanced luxury seats add motors for lumbar support, side bolsters, and even extendable thigh supports. Other applications include motors for **power folding mirrors**, **sunroofs**, and **power liftgates** or trunks. In all these cases, the DC motor provides the necessary torque and continuous motion to perform the task reliably over the life of the vehicle [88, 93].

Medium Stepper Motors (50 - 200 Watts)

The category of medium stepper motors, with power ratings specified between 50 and 200 watts, represents a more specialized and less commonly deployed segment within the landscape of automotive auxiliary motors. While small stepper motors are ubiquitous for low-power precision control and medium DC motors are the clear choice for continuous-duty tasks like pumping and ventilation, applications that require the specific combination of high precision and the moderate torque offered by a medium-power stepper motor are not prevalent in the common auxiliary systems examined for this report. The primary advantage of a stepper motor—its ability to provide precise angular control in an open-loop configuration—is most pronounced and cost-effective in smaller-scale applications where the load is predictable and relatively light. As power and torque requirements increase into the 50-200 watt range, the complexity and cost of the required driving electronics for stepper motors also rise substantially. In this power class, brushless DC motors paired with feedback sensors (in a closed-loop control system) often become a more practical, efficient, and robust solution for tasks that require both power and precision.

Despite their infrequent use in mainstream systems, there are theoretical and niche areas where medium stepper motors could be applied in advanced BEV systems. One potential area could be in sophisticated, high-load active aerodynamic systems. While current active grille shutters use low-power actuators, future systems featuring larger, faster-moving aerodynamic elements—such as large active spoilers or complex underbody diffusers that require very precise positioning for optimal downforce or drag reduction at various speeds—could potentially leverage the capabilities of a medium-power stepper motor. Another hypothetical application could be within certain advanced chassis systems, such as steer-by-wire or active roll stabilization, where precise and forceful actuation is needed. However, these safety-critical systems typically employ highly redundant and robust brushless DC motors with closed-loop feedback to ensure maximum reliability and fail-safe operation. Based on the available research data covering common auxiliary systems in 2024-2026 model year BEVs, there is no widespread, clearly identified application that consistently utilizes stepper motors in the 50-200 watt power range. The vast majority of non-drivetrain motor applications appear to be well-served by the established division of labor: sub-50-watt stepper motors for low-power precision tasks and 50-500 watt DC/BLDC motors for continuous fluid pumping, fan operation, and other convenience features. Therefore, this motor category remains largely specialized, and its deployment in the European BEV market within the specified timeframe is not prominently documented in the reviewed sources.

4. Motor Distribution Across Vehicle Segments

The quantity and complexity of auxiliary electric motors in a battery electric vehicle are not uniform across the market. Instead, there is a distinct and predictable trend of increasing motorization that correlates directly with a vehicle's size, price, and luxury positioning. As manufacturers seek to differentiate their products and add value, they incorporate a growing number of electrically powered comfort, convenience, and performance features, each requiring one or more dedicated motors. This section analyzes the distribution of the three defined motor categories across the A to F vehicle segments, providing detailed estimates for motor counts in various subsystems and a summary of the total motor content per segment. The analysis highlights the stark contrast between the minimalist approach in entry-level vehicles and the extensive electrification of features in premium and luxury models.

The following table, **TABLE 1**, provides a comprehensive breakdown of estimated motor counts for specific applications across the full range of vehicle segments. These figures are estimates based on typical feature sets for representative models within each class.

TABLE 1: Comprehensive Auxiliary Motor Distribution by System and Vehicle Segment (2024-2026)

System/ Applica- tion	Motor Type	A-seg- ment	B-seg- ment	C-seg- ment	D-seg- ment	E-seg- ment	F-seg- ment
HVAC System	Small Stepper	4-6	4-6	6-8	6-8	8-12	10-15
Head- lamp System	Small Stepper	2	2	2-4	4	4-6	6-8
Active Grille Shutter	Small Stepper	0-1	1-2	1-2	2	2-4	4-6
Char- ging Port Lock(s)	Small Stepper	1	1	1	1-2	2	2
Auto- matic/ Powere d Doors	Small Stepper	0	0	0	0	0-4	4
Window Lifts	Medium DC	2	4	4	4	4	4
Wind- shield Wipers	Medium DC	1-2	1-2	1-2	2	2	2
Coolant Pumps	Medium DC	1	1	1-2	2	2-3	3-4
Power Folding Mirrors	Medium DC	0	2	2	2	2	2
Power Seats (per seat)	Medium DC	0	0	2-4	4-6	5-8	8-12
Total Power Seats	Medium DC	0	0	4-8	8-12	10-16	16-24
		0	0	1-2	2	2	2-4

System/ Applica- tion	Motor Type	A-seg- ment	B-seg- ment	C-seg- ment	D-seg- ment	E-seg- ment	F-seg- ment
Sunroof/ Panor- amic Roof	Medium DC						
Power Liftgate/ Trunk	Medium DC	0	0	0	2	2	2
Soft- Close Doors	Medium DC	0	0	0	0-4	4	4
Power Steering Column	Medium DC	0	0	0	0	2	2
Deploy- able Door Handles	Medium DC	0	0	0	0	4	4

To provide a clearer picture of the overall trend, **TABLE 2** summarizes the total estimated motor counts for each category and vehicle segment, aggregating the detailed data from TABLE 1.

TABLE 2: Summary of Total Estimated Auxiliary Motor Counts per Vehicle Segment

Segment	Small Stepper Motors	Medium DC Motors	Medium Step- per Motors	Total Motors (Range)
A-segment	6-8	3-5	0	9-13
B-segment	8-10	6-8	0	14-18
C-segment	10-14	10-16	0	20-30
D-segment	12-18	15-25	0	27-43
E-segment	15-25	25-40	0	40-65
F-segment	20-30+	35-55+	0	55-85+

Analysis of Differences Between Vehicle Segments

The data presented in the tables reveals a clear and dramatic escalation in the number of auxiliary motors as one moves from the entry-level A-segment to the flagship F-segment. This proliferation is not arbitrary; it is a direct reflection of the increasing complexity and number of features offered in more expensive vehicles. The analysis of these differences can be broken down by segment tiers.

In the **A-segment and B-segment**, vehicles are engineered with a primary focus on cost-effectiveness, efficiency, and practicality. Consequently, their auxiliary motor count is the lowest. Motorization is generally restricted to essential functions. A-segment vehicles, such as the Fiat 500e, typically feature basic HVAC systems with a moderate number of stepper motor actuators (4-6), mandatory automatic headlamp leveling (2 steppers), and a motorized charging port lock. For medium DC motors, they are typically limited to the two front power windows, the windshield wipers, and a single essential coolant pump for thermal management. The B-segment, represented by models like the Peugeot e-208, expands on this slightly by making four-door power windows and power-folding mirrors common features, adding several more medium DC motors. Some B-segment models also introduce active grille shutters for improved aerodynamics, adding one or two small stepper motors. The total motor count in these segments ranges from approximately 9 to 18 units, establishing a baseline of essential electrification.

The transition to the **C-segment and D-segment** marks a significant step-up in comfort and convenience features, resulting in a substantial increase in motor count. C-segment vehicles like the Tesla Model 3 or Nissan Ariya often introduce dual-zone climate control, which requires more HVAC stepper motors (6-8) for independent temperature and airflow control for the driver and passenger. More advanced headlamp systems, such as matrix LEDs, may use additional stepper motors for more granular beam control. This is also the segment where basic power seats first appear as an option, adding four to eight medium DC motors for 4-way or 6-way adjustment on the front seats. The D-segment, including premium models like the BMW i4, builds upon this by making more advanced features standard. HVAC systems become more complex, adaptive headlamps are common, and power seats offer more axes of adjustment (e.g., 12-way or 16-way), which can double the number of seat motors. Features like a power liftgate (2 DC motors) and more sophisticated thermal management systems with multiple coolant pumps also become prevalent. The total motor count in these mid-range segments typically falls between 20 and 43 units, representing a doubling or even tripling of the count found in entry-level cars.

The pinnacle of auxiliary motor integration is found in the **E-segment and F-segment**, which encompass the executive and luxury classes. Vehicles like the Mercedes EQS and Porsche Taycan are showcases for automotive technology, and nearly every conceivable convenience feature is motorized. The number of small stepper motors grows significantly due to multi-zone (three, four, or even five-zone) climate control systems, which can require a dozen or more individual actuators to manage airflow and temperature throughout the entire cabin. Advanced active aerodynamic systems with multiple moving elements and highly complex adaptive headlamps also contribute to this increase. The explosion in the number of medium DC motors is even more pronounced. Seats in this class are not merely adjustable; they are often “executive” seats with heating, ventilation (requiring small internal fans), and complex multi-cell massage functions, each cell potentially driven by its own small motor or pneumatic actuator controlled by motorized valves. A single front seat in an F-segment vehicle can contain more motors than an entire A-segment car. Additional features like automatic or soft-closing doors, power-deployable door handles, power-adjustable steering columns, and expansive panoramic roofs with multiple moving shades all add to the tally. The total motor count in these top-tier segments can range from 40 to well over 85 motors, illustrating a philosophy where electrification is used to create an environment of ultimate comfort, convenience, and technological sophistication. The medium

stepper motor category remains absent, as even high-force precision tasks in these segments are typically handled by geared DC motors or other specialized actuators.

5. Material Composition and Technical Specifications

The performance, efficiency, durability, and cost of auxiliary electric motors are fundamentally determined by the materials from which they are constructed. The selection of specific metals, alloys, plastics, and magnetic materials is a complex engineering trade-off, balancing electromagnetic properties, thermal conductivity, mechanical strength, weight, and raw material cost. This section provides a detailed analysis of the material composition, weight ranges, and key technical specifications for the three defined motor categories. It also delves into the cost structures and the critical supply chain dynamics that influence the production of these essential BEV components.

TABLE 3: Estimated Material Composition and Specifications for Small Stepper Motors (<50W)

Component	Material	Percentage of Weight (Est.)	Specific Al-loys / Grades	Key Function / Properties
Stator Core	Electrical Steel	30-40%	Non-Grain Oriented (NOES) Silicon Steel (M19, M27, M36)	Provides a low-reluctance path for magnetic flux; laminations (0.20-0.35mm) reduce eddy current losses [21, 22, 23, 25].
Windings	Copper	9-10%	High-Purity Copper Wire	Conducts electrical current to generate electromagnetic fields; high conductivity minimizes resistive losses [13, 15, 16].
Rotor Magnet	NdFeB / Ferrite	15-25%	Sintered NdFeB (N42, N48SH, N45UH); High-Retentive Steel	Creates the permanent magnetic field for the rotor; Nd-FeB for high torque, high-temp grades (SH, UH) for thermal stability [28, 29, 31].
Housing/Frame	Aluminum / Plastic	15-20%	Die-Cast Aluminum; High-Strength Engineering Plastics	Provides structural support, protection from environment, and heat dissipation (for aluminum).
Shaft/Bearings	Steel	10-15%	Carbon Steel; Stainless Steel	Transmits mechanical motion; provides low-friction rotation.

TABLE 4: Estimated Material Composition and Specifications for Medium DC Motors (50-500W)

Component	Material	Percentage of Weight (Est.)	Specific Al- loys / Grades	Key Function / Properties
Housing/Frame	Aluminum / Steel	35-45%	Die-Cast Alumin- um Alloys; Fab- ricated Steel; Cast Iron (rare)	Provides robust structural sup- port and acts as a heat sink to dissipate thermal energy from windings [35, 36, 37, 53, 55].
Windings	Copper	15-18%	100% High-Pur- ity Copper Wire	Carries higher currents for greater power output; high thermal con- ductivity helps manage heat [13, 15].
Magnets (Stator/Rotor)	Ferrite / NdFeB	20-30%	Strontium/Bari- um Ferrite; Sintered NdFeB (e.g., N-series, SH/UH grades)	Creates the magnetic field. Ferrite is low- cost and thermally stable; NdFeB provides high power density [60, 61, 65].
Shaft	Steel	10-15%	High-Strength Carbon Steel (SAE 1045); Al- loy Steel (4140)	Transmits higher torque and with- stands greater mechanical stress [57, 58].
Brushes/Com- mutator	Carbon / Copper	<5% (Brushed only)	Graphite-Copper Composite; Cop- per Alloy	Provides mechanical commutation of current to the rotor windings in brushed DC mo- tors [35].

TABLE 5: Estimated Material Composition and Specifications for Medium Stepper Motors (50-200W)

Component	Material	Percentage of Weight (Est.)	Specific Al-loys / Grades	Key Function / Properties
Stator Core	Electrical Steel	35-45%	High-Grade NOES (M27, M36); Thin Gauges (<0.35mm)	Manages higher magnetic flux density and minimizes core losses at higher power levels [21, 22, 25].
Windings	Copper	12-15%	High-Purity Cop-per Wire	Larger gauge wire to handle increased cur-rent for higher torque output [68, 70, 71].
Rotor Magnet	NdFeB	25-35%	High-Coercivity Sintered NdFeB (48SH, 45UH, 42EH)	Provides very high magnetic energy product for maximum torque; high-temp grades are essential to pre-vent demagnet-ization [29].
Housing/Frame	Aluminum / Steel	15-25%	Die-Cast Alumin-um; Steel	Provides rigid structural sup-port and effect-ive heat management for higher thermal loads.
Gearbox (Op-tional)	Steel / Powder Metal	Varies	Hardened Alloy Steel; Powder Metallurgy Com-ponents	Multiplies torque output; alloy steel for high precision and durability [70, 71].

TABLE 6: Comparative Overview of Auxiliary Motor Weight Ranges and Specifications

Attribute	Small Stepper Motor	Medium DC Motor	Medium Stepper Motor
Power Range	< 50 Watts	50 - 500 Watts	50 - 200 Watts
Typical Weight Range	0.2 - 0.4 kg [7, 8, 10, 11]	2.3 - 3.6 kg (for 200W) [41, 42, 43, 44, 45]	0.7 - 1.5+ kg
Primary Function	Precision Positioning	Continuous Motion/Torque	High-Torque Positioning
Control Method	Open-Loop (Step-based)	Closed-Loop (Voltage/PWM)	Open-Loop (Step-based)
Key Advantage	High precision, low cost, simple control	High power/torque, reliability (BLDC)	High holding torque, precision
Common Magnet Type	Hybrid (NdFeB), PM (Ferrite) [1, 3, 4, 28]	BLDC (NdFeB), Brushed (Ferrite) [35, 63, 65]	Hybrid (High-Grade NdFeB) [68, 69]
Typical Application	HVAC Actuators, Headlamp Leveling [3, 6]	Coolant Pumps, Window Lifts, Seats [88, 93]	Niche/Specialized Actuation

Cost Analysis and Supply Chain Information

The cost structure of auxiliary electric motors is a complex equation determined by raw material prices, manufacturing complexity, labor costs, and production volume. The most significant and volatile cost drivers across all high-performance motor types are **copper** and **rare earth elements**. Copper, which is essential for the windings in all three motor categories, constitutes a significant portion of the motor’s weight—up to 18% in medium DC motors [13, 15]. The global price of copper is subject to considerable fluctuation, driven by a tightening supply-demand balance as global electrification efforts in transportation, energy generation, and infrastructure accelerate. This sustained upward pressure on copper prices directly impacts the base manufacturing cost of every motor.

Even more critical to the cost of high-performance stepper and BLDC motors is the price of **Neodymium Iron Boron (NdFeB) permanent magnets**. For a technologically analogous brushless motor, historical analysis has shown that the magnet material can account for over 70% of the total material cost [74]. This disproportionate impact means that motor cost is acutely sensitive to the volatile market for rare earth elements. The price of neodymium, and particularly the heavy rare earths like dysprosium and terbium that are added to create high-temperature-tolerant magnets, can fluctuate dramatically based on market speculation and geopolitical factors. The higher the required thermal performance (e.g., moving from an ‘H’ grade to a ‘UH’ or ‘EH’ grade magnet), the higher the cost, making the selection of magnet grade a critical factor in the final price of the motor.

This leads to the single most significant challenge in the supply chain for auxiliary motors: a profound and systemic dependence on China. As of 2025, China controls an estimated 70% of global rare earth mining, 85% of the refining and processing, and approximately 90% of the global production of finished NdFeB magnets [81]. This near-monopoly creates a critical single-point-of-failure risk for the entire global automotive industry, which is increasingly reliant on these magnets for dozens of auxiliary motors per vehicle, in addition to the main traction motor. Any change in Chinese industrial or export policy, such as the implementation of export licensing for magnet materials and technologies, can result in immediate and severe price shocks and supply disruptions for automakers worldwide [81].

In response to this critical vulnerability, a multi-pronged strategic realignment is underway across the industry and at governmental levels. Automakers and Tier 1 suppliers are actively pursuing **supply chain diversification**, seeking to establish contracts with and support the growth of non-Chinese magnet producers, even if it means paying a premium for materials [81]. A second strategy is **vertical integration and upstream investment**. Major automakers like General Motors have formed strategic partnerships with mining companies such as MP Materials to help establish a complete mine-to-magnet supply chain within the United States [81]. This is complemented by government initiatives like the U.S. Inflation Reduction Act and Europe's Critical Raw Materials Act, which provide financial incentives and policy support to foster domestic mining, processing, and manufacturing of these critical materials [81]. Finally, extensive research and development is focused on **technology substitution**. This includes efforts to design high-performance magnets that reduce or eliminate the need for the most critical heavy rare earths, as well as developing entirely new rare-earth-free motor architectures [81]. However, for the foreseeable future, NdFeB magnets are expected to remain the dominant technology for performance-critical applications, making the management of their fragile supply chain a top strategic priority for the automotive sector.

6. Estimates and Methodology

Transparency regarding the origin and nature of the data presented in this report is essential for its proper interpretation. It is critical to distinguish between data that is documented and verifiable from established sources, and data that represents reasoned estimates based on industry norms and logical inference. This section provides a clear explanation of the methodology used to derive the motor count estimates and outlines the key assumptions and limitations inherent in this analysis.

A significant portion of the quantitative data in this report, particularly the motor counts presented in Section 4 (TABLE 1 and TABLE 2), are **estimates**. These figures are not the result of a physical tear-down of every representative vehicle model. Instead, they were compiled through a systematic process of analyzing the documented feature sets of typical vehicles within each market segment (A through F). The basis for these estimates is the public and technical information released by automotive manufacturers for 2024-2026 models, combined with established knowledge of how these features are typically implemented from an engineering perspective.

The methodology for generating these estimates can be described as a bottom-up synthesis. For each vehicle segment, a set of representative models was identified (e.g., Fiat 500e for A-segment, BMW i4 for D-segment). The standard and optional features for these models were then reviewed. For each feature known to be motorized (e.g., "power windows," "dual-zone climate control," "adaptive headlamps"), a corresponding number of motors was assigned based on common industry practice. For example:

- * A "power windows" feature on a four-door car is assigned four medium DC motors.
- * A "dual-zone" climate control system is estimated to require more stepper motor actuators (e.g., 6-8) than a basic single-zone system (e.g., 4-6) to control the additional blend doors.
- * An "adaptive headlamp" system is assigned four small stepper motors (one for vertical and one for

horizontal adjustment in each of the two headlamps), whereas a basic “automatic leveling” system is assigned only two (one vertical motor per headlamp).

* A “12-way power seat” is estimated to require six medium DC motors (fore/aft, front height, rear height, recline, lumbar in/out, lumbar up/down).

The ranges provided in the tables (e.g., “9-13 motors”) reflect the variability in standard equipment versus optional upgrades within a segment. The lower end of the range represents a base model, while the higher end represents a well-equipped or “fully loaded” model within that same segment. The assumption that medium stepper motors (50-200W) have a count of zero across all segments is based on the lack of documented, widespread applications for this motor type in the common auxiliary systems reviewed. Their use is likely confined to proprietary or highly specialized systems not covered by general technical literature.

This estimation methodology is subject to several **limitations and assumptions**. The primary limitation is the reliance on publicly available information, which can vary in its level of technical detail. The exact number of motors in a complex system like a multi-zone HVAC or a massaging seat can be proprietary information and may differ between manufacturers. The analysis assumes a degree of uniformity in engineering solutions across the industry, which may not always be the case. Furthermore, the motor counts represent a “typical” vehicle in the segment, and actual counts can vary significantly between different manufacturers and even between different trim levels of the same model. The classification of motors is also an assumption; while a window lift is almost certainly a medium DC motor, the specific choice between a small stepper and a small DC motor for a function like a charging port lock might vary. Despite these limitations, the methodology provides a robust and directionally accurate framework for understanding the scale of motorization across the BEV market and for illustrating the powerful trend of increasing feature complexity and motor content in higher vehicle segments.

7. Conclusions

This comprehensive investigation into the auxiliary electric motors of European and Asian BEVs for the 2024-2026 model years reveals a complex and rapidly evolving ecosystem of components that are fundamental to modern vehicle function. The analysis confirms that the transition to electric mobility has catalyzed a massive proliferation of non-drivetrain motors, turning vehicles into sophisticated mechatronic systems. The findings of this report offer several key conclusions regarding the roles of different motor types, their distribution across the market, and the critical industry trends and strategic implications that will shape the future.

First, there is a clear and specialized division of labor among motor types. **Small stepper motors (<50W)** are the undisputed choice for applications demanding high-precision, low-power positioning. Their standard fitment in HVAC actuators and adaptive headlamp systems is a testament to their cost-effectiveness and reliability in open-loop control scenarios [1, 3, 6]. Their role is crucial for enabling the advanced climate control and safety lighting features that consumers now expect. In contrast, **medium DC motors (50-500W)**, particularly brushless (BLDC) variants, are the indispensable workhorses for continuous-duty applications. Their most critical function is driving the electric pumps for vehicle thermal management, a task absolutely essential for battery health, performance, and safety [88, 93]. The efficiency and controllability of these motors are paramount, as their energy consumption is a direct variable in the vehicle’s overall driving range. The category of **medium stepper motors (50-200W)** was found to be a niche segment with no broadly identified applications in common systems, indicating that the current needs of the market are well-served by the combination of small steppers and medium DC motors.

Second, the report definitively establishes a direct and exponential correlation between a vehicle's market segment and its total auxiliary motor count. The journey from an A-segment mini car, with as few as nine motors, to an F-segment luxury sedan, with upwards of eighty-five, is a story of escalating feature content and electrification. This trend is driven by the competitive need to offer greater comfort, convenience, and technological sophistication in premium and luxury models. Features that were once mechanical or non-existent—from seat adjustments and door handles to multi-zone climate control and active aerodynamics—are now almost universally motorized in high-end vehicles. This proliferation has significant implications for vehicle electrical architecture, power distribution, network bandwidth, and overall cost and complexity.

Finally, the analysis of material composition and supply chains highlights the most significant strategic challenges facing the industry. The performance of modern auxiliary motors is heavily reliant on advanced materials, including high-grade electrical steel, high-purity copper, and, most importantly, high-strength **Neodymium Iron Boron (NdFeB) rare earth magnets** [28, 60, 74]. While these materials enable the creation of compact, efficient, and powerful motors, they also introduce significant cost volatility and supply chain risk. The price of copper continues to be a concern, but the industry's overwhelming dependence on China for the mining, processing, and manufacturing of NdFeB magnets represents a systemic vulnerability [81]. This geopolitical risk has become a primary focus for automakers and governments alike, spurring a strategic imperative to de-risk the supply chain through diversification, investment in domestic production, and research into alternative technologies [81, 82, 84]. The future trajectory of BEV development, particularly regarding cost reduction and production stability, will be inextricably linked to the successful navigation of these material and supply chain challenges. In essence, the humble auxiliary motor has become a microcosm of the larger technological, economic, and geopolitical forces shaping the future of mobility.

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